

Task 1: Three common hacking methodologies

Methodology	How it works	Example involving Instagram/Paytm
Phishing	The attacker creates a fake login page or sends a deceptive message to trick the victim into entering credentials.	An attacker sends an Instagram user a fake "Your account will be suspended" message containing a fake login link. If the victim enters their username and password, the credentials can be stolen.
Brute Force	The attacker systematically tries many possible passwords or PINs until the correct one is found.	If an account has a weak 4-digit PIN and there are no effective rate limits, an attacker could repeatedly try possible PINs until the correct one is discovered.
SQL Injection	An attacker manipulates poorly protected database queries through application input fields to access or alter database information.	A vulnerable login form could potentially allow an attacker to manipulate a database query and bypass authentication or access unauthorized records.

Task 2: Python brute-force simulation

This is a local educational simulation. It does not connect to Instagram, Paytm, or any other service.

Simple 4-digit PIN brute-force simulation

```
preset_pin = "5729"  
attempts = 0
```

```
for number in range(10000):  
    pin = f"{number:04d}"
```

```
attempts += 1
```

```
if pin == preset_pin:  
    print("PIN found:", pin)  
    print("Attempts taken:", attempts)  
    break
```

How it works

- preset_pin contains the PIN we want to find.
- range(10000) generates numbers from 0 to 9999.
- f"{number:04d}" converts them into 4-digit format:
 - 0000
 - 0001
 - 0002
 - ...
 - 9999
- attempts counts how many combinations were tested.
- When the generated PIN equals the preset PIN, the loop stops.

For 5729, the output will be:

PIN found: 5729

Attempts taken: 5730

This demonstrates why short PINs are weak when an authentication system doesn't enforce protections such as rate limiting or account lockout.

Task 3: Phishing attack targeting a food-delivery app

Step 1 — Creating the fake message

Attacker: Creates a convincing message pretending to be from the food-delivery company. For example, it might claim that the user's account has an unusual login or that a refund requires verification.

Victim: Receives the message and may believe it is an official communication.

Step 2 — Sending the phishing link

Attacker: Sends a link directing the victim to a fraudulent website designed to resemble the legitimate service.

Victim: Clicks the link because the message appears urgent or genuine.

Step 3 — Credential harvesting

Attacker: The fake website asks for information such as a username, phone number, password or OTP. Information entered into the fraudulent page can be captured by the attacker.

Victim: Enters their information believing they are logging into the legitimate service.

Step 4 — Misuse of stolen information

Attacker: Attempts to use the captured information for unauthorized access, account takeover, impersonation or further fraud.

Victim: May lose access to the account or have personal information misused.

Task 4: Fake login page — two credential-harvesting methods

Method 1 — Sending credentials to an attacker-controlled server

The HTML form can be configured so that when the victim clicks Login, the entered username and password are submitted to a server controlled by the attacker.

For example, conceptually:

- `<form action="attacker-controlled-server" method="POST">`

The victim thinks they are logging into the legitimate service, but the submitted information is instead sent elsewhere.

Method 2 — Storing or processing credentials through malicious page code

A fraudulent page may use client-side JavaScript or another mechanism to capture information entered into the form and transmit it to an attacker-controlled location.

For example, the attacker could capture:

- Username/email
- Password
- Phone number
- Other information entered into the form

The page could then redirect the victim to the legitimate website afterward, making the attack less obvious.

Security lesson: A login page should only send credentials to a trusted HTTPS endpoint, and developers should use proper authentication systems rather than collecting passwords through an untrusted form.

Task 5: Five real-world Indian hacking incidents

Incident	Year	Methodology
KiranaPro cyberattack	2025	Unauthorized access led to company

		servers/data being wiped. The company initially blamed a former employee, while later reporting said it could not rule out an external hack.
Zomato data breach	2017	Compromised developer credentials were used to access GitHub; the attacker then exploited a vulnerability in old application code to access database information.
Zomato security breach	2015	A security researcher gained unauthorized access to Zomato user information through a website vulnerability.
Indian app-based communication services targeted by impersonation/phishing scams	2025	Phishing, impersonation and account misuse were significant enough for the Department of Telecommunications to introduce SIM-binding requirements for certain app-based communication services.
KiranaPro former-employee account incident	2025	Possible misuse of an account belonging to a former employee whose access had reportedly not been deactivated; the company said a full forensic investigation was needed to establish exactly what happened.